

Project Name	CarPro Rental Management System	Creator	Nguyễn Ngọc Anh, Bùi Xuân Bắc, Trần Thị Minh Tâm, Nguyễn Lâm Tùng, Hoàng Ngọc Tĩnh
Project Code	SWP391	Issue Date	21/07/2026
Document Code	SWP391_Test Report	Version	v1.2 (Round 2 executed)

Record of change

Effective Date	Version	Change Item	*A,D,M	Change description	Reference
10/06/2026	v0.1	System test for assigned	A/M	Create a detailed individual System Test report based on	Project Tracking / UC Specification / UI-UX
10/06/2026	v0.2	Individual system test for	A/M	Create a detailed individual System Test report based on	Project Tracking / UC Specification / UI-UX
10/06/2026	v0.2	Individual System Test Report	A/M	Create an individual System Test Report for the assigned	Project Tracking / UC Specification / UI-UX.
10/06/2026	v0.2	Individual system test for	A/M	Create a detailed individual System Test report based on	Project Tracking / UC Specification / UI-UX
23/06/2026	v0.3	Update individual system test	M	Add and update detailed test cases for Contract and	Project Tracking / UC Specification / UI-UX
10/06/2026	v0.2	Individual system test for	A/M	Create a detailed individual System Test report based on	Project Tracking / UC Specification / UI-UX.
21/07/2026	v1.0	Merge 5 individual reports	M	Combine Cover, Test Cases, Test Statistics, all 10 workflow	Merged from 5 individual SWP391_Test Report files
22/07/2026	v1.2	Round 2 test execution after fixing test setup/data issues	M	Re-ran 8 test cases that failed in Round 1 due to test setup/data problems (shared test vehicles/dates, missing required fields, wrong test method) using isolated data and correct API calls. All 8 now Pass. Confirmed 9 real code defects remain (incl. 1 newly discovered: ContractService does not block contract creation for unverified customers).	Round 2 automated test run (team members)

TEST CASE LIST

Project Name	<i>CarPro Rental Management System</i>
Project Code	<i>SWP391</i>
Test Environment Setup Description	<i>1. Server: local development environment for Java Servlet/JSP application.</i> <i>2. Database: SQL Server/SSMS test database with sample accounts, vehicles, bookings, contracts and payments.</i> <i>3. Web Browser: Chrome/Edge browser.</i> <i>4. Test method: execute UI workflow according to each test case and collect screenshot evidence.</i> <i>5. Test scope: member-owned functions only, including positive, negative, validation and permission cases.</i>

No	Function Name	Sheet Name	Description	Pre-Condition
1	Register a new customer account with valid information.	Auth and Profile Workflow	Register a new customer account with valid information.	Guest is not logged in; email and phone are not duplicated.
2	Register with an email that already exists.	Auth and Profile Workflow	Register with an email that already exists.	A customer account with the same email already exists.
3	Register with missing required fields.	Auth and Profile Workflow	Register with missing required fields.	Guest is on Register screen.
4	Register with invalid information.	Auth and Profile Workflow	Register with invalid information.	Guest is on Register screen.
5	Register with password and confirm password mismatch.	Auth and Profile Workflow	Register with password and confirm password mismatch.	Guest is on Register screen.
6	Log in successfully with a valid customer account.	Auth and Profile Workflow	Log in successfully with a valid customer account.	A valid active customer account exists.
7	Log in with invalid information.	Auth and Profile Workflow	Log in with invalid information.	Customer is on Login screen.
8	Log in with wrong password.	Auth and Profile Workflow	Log in with wrong password.	A valid account exists.
9	Log in with a locked account.	Auth and Profile Workflow	Log in with a locked account.	The selected account is locked by Admin/Staff.
10	Login with missing required fields.	Auth and Profile Workflow	Login with missing required fields.	Customer is on Login screen.
11	Log out from the system.	Auth and Profile Workflow	Log out from the system.	User is logged in.
12	Update customer profile with valid personal information.	Auth and Profile Workflow	Update customer profile with valid personal information.	Customer is logged in.
13	Update profile with invalid phone number format.	Auth and Profile Workflow	Update profile with invalid phone number format.	Customer is logged in.
14	Verify customer profile completeness before booking.	Auth and Profile Workflow	Verify customer profile completeness before booking.	Customer profile contains required ID and driver license information.
15	Reject customer profile verification when required document information is missing.	Auth and Profile Workflow	Reject customer profile verification when required document information is missing.	Customer profile is missing ID or driver license information.
16	Admin views the user list.	User Management Workflow	Admin views the user list.	Admin account exists and user data exists.
17	Search user by keyword.	User Management Workflow	Search user by keyword.	Admin/Staff is logged in; matching data exists.
18	Filter users by role or status.	User Management Workflow	Filter users by role or status.	User data contains multiple roles/statuses.
19	Update user account information successfully.	User Management Workflow	Update user account information successfully.	Admin is logged in; target account exists.
20	Lock a user account.	User Management Workflow	Lock a user account.	Target user is active.
21	Unlock a user account.	User Management Workflow	Unlock a user account.	Target user is locked.
22	Check availability for a vehicle with	Booking Customer Workflow	Check availability for a vehicle with valid rental dates.	Vehicle is active and has no overlapping approved booking.
23	Create a booking request	Booking Customer Workflow	Create a booking request successfully.	Customer has completed required profile information; vehicle is available.
24	Create booking and verify default	Booking Customer Workflow	Create booking and verify default status.	Booking creation succeeds.
25	View booking list as Customer.	Booking Customer Workflow	View booking list as Customer.	Customer has at least one booking.
26	View booking detail.	Booking Customer Workflow	View booking detail.	Selected booking exists.
27	Check availability when vehicle is	Booking Customer Workflow	Check availability when vehicle is already booked in the selected period.	Vehicle has an existing approved booking in the same period.
28	Check availability with return date	Booking Customer Workflow	Check availability with return date before rental date.	Customer is logged in.
29	Check availability without logging	Booking Customer Workflow	Check availability without logging in.	Guest is not logged in.
30	Reject booking when required	Booking Customer Workflow	Reject booking when required booking information is missing.	Customer is logged in.
31	Reject booking when customer	Booking Customer Workflow	Reject booking when customer profile is not verified/complete.	Customer profile is incomplete or unverified.
32	Update a pending booking with	Booking Customer Workflow	Update a pending booking with valid new dates.	Booking status is Pending and vehicle is available for new dates.
33	Reject update after booking is	Booking Customer Workflow	Reject update after booking is approved or handed over.	Booking status is Confirmed, Contracted or Handover/Returned.
34	Cancel a pending booking.	Booking Customer Workflow	Cancel a pending booking.	Booking status is Pending.
35	Reject cancellation when booking is	Booking Customer Workflow	Reject cancellation when booking is already handed over.	Booking is in handover/rental process.
36	View booking calendar as	Booking Staff Workflow	View booking calendar as Staff/Admin.	Staff/Admin is logged in; booking data exists.
37	Filter booking calendar by	Booking Staff Workflow	Filter booking calendar by vehicle/status/date range.	Booking data exists.
38	View booking policy before	Booking Staff Workflow	View booking policy before processing booking.	Booking policy data exists.
39	Staff approves a valid pending	Booking Staff Workflow	Staff approves a valid pending booking.	Pending booking exists; no schedule conflict; customer profile is valid.
40	Staff rejects a booking with a	Booking Staff Workflow	Staff rejects a booking with a reason.	Pending booking exists.
41	Reject approving booking when	Booking Staff Workflow	Reject approving booking when vehicle is no longer available.	Pending booking overlaps with an already approved booking.
42	Prevent unauthorized user from	Booking Staff Workflow	Prevent unauthorized user from processing bookings.	Customer account exists.
43	Process vehicle handover	Handover Return Workflow	Process vehicle handover successfully.	Approved contract exists; deposit requirement is satisfied according to policy.
44	Reject handover when deposit is not	Handover Return Workflow	Reject handover when deposit is not paid.	Approved booking exists but deposit is unpaid.

45	Reject handover with missing	Handover Return Workflow	Reject handover with missing odometer/fuel/condition information.	Staff is logged in; approved contract exists.
46	Prevent duplicate handover for the	Handover Return Workflow	Prevent duplicate handover for the same booking.	Booking is already handed over.
47	Process vehicle return with full	Handover Return Workflow	Process vehicle return with full information.	Vehicle is currently handed over/active rental.
48	Record returned vehicle with	Handover Return Workflow	Record returned vehicle with damage note.	Vehicle is currently handed over/active rental.
49	Reject return when return odometer	Handover Return Workflow	Reject return when return odometer is lower than handover odometer.	Handover odometer exists.
50	Settle deposit with no extra charge.	Handover Return Workflow	Settle deposit with no extra charge.	Vehicle return completed; no extra fee exists.
51	Settle deposit with deducted	Handover Return Workflow	Settle deposit with deducted additional fee.	Return completed and additional fee exists.
52	Reject settlement before vehicle	Handover Return Workflow	Reject settlement before vehicle return is completed.	Booking is handed over but not returned.
53	Record an additional fee after	Report Fee Workflow	Record an additional fee after return.	Returned booking exists.
54	Reject additional fee with invalid	Report Fee Workflow	Reject additional fee with invalid amount.	Staff/Admin is logged in.
55	Update additional fee reason before	Report Fee Workflow	Update additional fee reason before settlement is completed.	Additional fee exists and settlement is not closed.
56	View revenue report by date range.	Report Fee Workflow	View revenue report by date range.	Payment and booking data exists.
57	Export or view revenue report	Report Fee Workflow	Export or view revenue report details.	Revenue report has data.
58	View vehicle utilization report.	Report Fee Workflow	View vehicle utilization report.	Booking and vehicle data exists.
59	Handle report with no data.	Report Fee Workflow	Handle report with no data.	No payment/booking records exist for selected range.
60	Prepare contract successfully from	Contract Payment Workflow	Prepare contract successfully from confirmed booking.	Booking status = CONFIRMED; customer verified; vehicle exists.
61	Reject contract creation when	Contract Payment Workflow	Reject contract creation when booking is not CONFIRMED.	Booking status ≠ CONFIRMED.
62	Reject contract creation when	Contract Payment Workflow	Reject contract creation when customer profile is not verified.	Booking confirmed; customer not verified.
63	Reject contract creation when	Contract Payment Workflow	Reject contract creation when vehicle does not exist.	Vehicle record missing.
64	View contract detail successfully.	Contract Payment Workflow	View contract detail successfully.	Contract exists.
65	Redirect unauthenticated user to	Contract Payment Workflow	Redirect unauthenticated user to login page.	User not logged in.
66	Reject viewing a non-existing	Contract Payment Workflow	Reject viewing a non-existing contract.	Invalid contract ID.
67	Deny viewing another customer's	Contract Payment Workflow	Deny viewing another customer's contract.	Customer does not own contract.
68	Update draft contract successfully.	Contract Payment Workflow	Update draft contract successfully.	Contract status = DRAFT.
69	Reject updating non-DRAFT	Contract Payment Workflow	Reject updating non-DRAFT contract.	Contract status ≠ DRAFT.
70	Prevent customer from updating	Contract Payment Workflow	Prevent customer from updating contract.	Customer account.
71	Reject invalid contract information.	Contract Payment Workflow	Reject invalid contract information.	Draft contract exists.
72	Complete electronic signature	Contract Payment Workflow	Complete electronic signature successfully.	Draft contract exists.
73	Reject signature when identity	Contract Payment Workflow	Reject signature when identity verification fails.	OTP verification enabled.
74	Record payment successfully.	Contract Payment Workflow	Record payment successfully.	Booking exists.
75	Reject payment with invalid	Contract Payment Workflow	Reject payment with invalid amount.	Booking exists.
76	Reject payment for non-existing	Contract Payment Workflow	Reject payment for non-existing booking.	Invalid booking ID.
77	Approve pending cash payment.	Contract Payment Workflow	Approve pending cash payment.	Pending cash payment exists.
78	Create bank transfer payment	Contract Payment Workflow	Create bank transfer payment successfully.	Bank transfer enabled.
79	Complete payment through	Contract Payment Workflow	Complete payment through webhook.	Pending payment exists.
80	Reject invalid webhook.	Contract Payment Workflow	Reject invalid webhook.	Pending payment exists.
81	Record refund successfully.	Contract Payment Workflow	Record refund successfully.	Refund amount exists.
82	Reject invalid refund amount.	Contract Payment Workflow	Reject invalid refund amount.	Refund transaction.
83	Export VAT invoice successfully.	Contract Payment Workflow	Export VAT invoice successfully.	Contract COMPLETED and fully paid.
84	Reject VAT export for unpaid	Contract Payment Workflow	Reject VAT export for unpaid contract.	Contract not fully paid.
85	Reject duplicate VAT invoice.	Contract Payment Workflow	Reject duplicate VAT invoice.	Active VAT invoice exists.
86	Update payment configuration	Configure Policy Workflow	Update payment configuration successfully.	Admin account.
87	Reject invalid payment	Configure Policy Workflow	Reject invalid payment configuration.	Admin account.
88	Create new rental policy	Configure Policy Workflow	Create new rental policy successfully.	Admin account.
89	Reject duplicate policy key.	Configure Policy Workflow	Reject duplicate policy key.	Existing policy key.
90	View vehicle catalog as Guest.	Vehicle Customer Workflow	View vehicle catalog as Guest.	Vehicle test data exists.
91	View vehicle catalog as Customer.	Vehicle Customer Workflow	View vehicle catalog as Customer.	Customer is logged in; vehicle data exists.
92	Handle empty vehicle catalog.	Vehicle Customer Workflow	Handle empty vehicle catalog.	No active vehicle matches the current query.
93	Search vehicle by keyword.	Vehicle Customer Workflow	Search vehicle by keyword.	Matching vehicle exists.
94	Filter vehicles by price range.	Vehicle Customer Workflow	Filter vehicles by price range.	Vehicle data contains different prices.
95	Filter vehicles by availability/status.	Vehicle Customer Workflow	Filter vehicles by availability/status.	Vehicle data contains multiple statuses.
96	Search with no matching result.	Vehicle Customer Workflow	Search with no matching result.	No vehicle matches the keyword.
97	View vehicle detail from catalog.	Vehicle Customer Workflow	View vehicle detail from catalog.	Selected vehicle exists.
98	Prevent booking action for	Vehicle Customer Workflow	Prevent booking action for unavailable vehicle.	Selected vehicle status is unavailable/rented/maintenance.
99	Staff/Admin views internal vehicle	Vehicle Admin Workflow	Staff/Admin views internal vehicle management list.	Staff/Admin account exists; vehicle data exists.
100	Create a new vehicle with valid	Vehicle Admin Workflow	Create a new vehicle with valid information.	Staff/Admin is logged in; plate number is unique.
101	Reject creating vehicle with	Vehicle Admin Workflow	Reject creating vehicle with duplicated plate number.	A vehicle with the same plate exists.
102	Update vehicle price and	Vehicle Admin Workflow	Update vehicle price and description.	Selected vehicle exists.
103	Customer accesses Vehicle	Vehicle Admin Workflow	Customer accesses Vehicle Management page directly.	Customer account exists and is logged in.
104	Create vehicle with empty required	Vehicle Admin Workflow	Create vehicle with empty required fields.	Admin/Staff is logged in.
105	Upload invalid image format.	Vehicle Admin Workflow	Upload invalid image format.	Admin/Staff is logged in.
106	Update vehicle with negative rental	Vehicle Admin Workflow	Update vehicle with negative rental price.	Vehicle exists.
107	Update vehicle status to	Vehicle Admin Workflow	Update vehicle status to Maintenance.	Vehicle is not in active handover/rental process.
108	Prevent setting a rented vehicle to	Vehicle Admin Workflow	Prevent setting a rented vehicle to Available manually.	Vehicle is rented/handed over.
109	Schedule maintenance for a vehicle.	Vehicle Admin Workflow	Schedule maintenance for a vehicle.	Staff/Admin is logged in; vehicle exists.
110	Reject maintenance schedule with	Vehicle Admin Workflow	Reject maintenance schedule with invalid date range.	Staff/Admin is logged in.

TEST STATISTICS

Project Name	CarPro Rental Management System	Creator	
Project Code	SWP391	Reviewer/ Approver	
Document Code	SWP391_Test Report	Issue Date	
Notes	Team system test report — 5 members (Anh, Bắc, Tâm, Tùng, Tĩnh), 10 modules, 110 test cases. Round 1 (22/07/2026)		

SWP391
Team (Anh,
Lecturer /
Reviewer
10/06/2026

No	Module code	Round 1				Round 2				Final Result				Number of test cases
		Passed	Failed	g	N/A	Passed	Failed	Pending	N/A	Passed	Failed	Pending	N/A	
1	Customer Account, Login and Profile Workflow	17	3	0	0	3	0	0	17	15	0	0	0	15
2	Internal User Account Management Workflow	6	0	0	0	0	0	0	6	6	0	0	0	6
3	Customer Booking Creation and Management Workflow	14	0	0	0	0	0	0	14	14	0	0	0	14
4	Staff Booking Review, Calendar and Policy Workflow	7	0	0	0	0	0	0	7	7	0	0	0	7
5	Vehicle Handover, Return Processing and Deposit Settlement	10	0	0	0	0	0	0	10	10	0	0	0	10
6	Additional Fee and Reporting Workflow	6	1	0	0	1	0	0	6	7	0	0	0	7
7	Contract Management and Payment Workflow	25	1	0	0	1	0	0	25	26	0	0	0	26
8	Policy Configuration Workflow	3	1	0	0	1	0	0	3	4	0	0	0	4
9	Vehicle Catalog, Search and Detail Workflow	8	1	0	0	1	0	0	8	9	0	0	0	9
10	Vehicle Management, Status and Maintenance Workflow	10	2	0	0	2	0	0	10	12	0	0	0	12
Sub total		106	9	0	0	9	0	0	106	110	0	0	0	110

Workflow	Customer Account, Login and Profile Workflow			
Test requirement	Covers customer access functions: register account, login account, logout, update profile and verify customer profile. Includes positive, negative and role-based validation cases.			
Number of TCs	13			
Testing Round	Passed	Failed	Pending	N/A
Round 1	17	3	0	0
Round 2	3	0	0	17
Final	15	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note
Scenario 1 - Customer Registration											
TC-ANH-AUTH-01	Register a new customer account with valid information.	1. Open Register screen. 2. Enter full name, email, phone, password and confirm password. 3. Click Register.	System creates a Customer account, shows success message and allows the user to go to Login screen.	Guest is not logged in; email and phone are not duplicated.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Register Success
TC-ANH-AUTH-02	Register with an email that already exists.	1. Open Register screen. 2. Enter a registered email with otherwise valid data. 3. Click Register.	System rejects registration and shows a clear duplicated email validation message. No new account is created.	A customer account with the same email already exists.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Register Success
TC-ANH-AUTH-03	Register with missing required fields.	1. Open Register screen. 2. Leave full name, email, phone or password empty. 3. Click Register.	System highlights required fields and does not submit the form.	Guest is on Register screen.	Failed	22/07/2020	Nguyễn Ngọc Anh	Passed	##	Nguyễn Ngọc Anh	Actual: Empty Inputs Allowed
TC-ANH-AUTH-04	Register with invalid information.	1. Open Register screen. 2. Enter email, phone or password invalid. 3. Click Register.	System shows invalid information message and keeps the user on Register screen.	Guest is on Register screen.	Failed	22/07/2020	Nguyễn Ngọc Anh	Passed	##	Nguyễn Ngọc Anh	Actual: Invalid Input Allowed
TC-ANH-AUTH-05	Register with password and confirm password mismatch.	1. Open Register screen. 2. Enter valid information. 3. Enter different values for Password and Confirm Password. 4. Click Register.	System shows password mismatch message and keeps the user on Register screen.	Guest is on Register screen.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Registration Rejected
Scenario 2 - Login and Logout											
TC-ANH-AUTH-06	Log in successfully with a valid customer account.	1. Open Login screen. 2. Enter valid email/username and password. 3. Click Login.	System authenticates successfully, creates session and redirects to the correct customer home page.	A valid active customer account exists.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Login Success
TC-ANH-AUTH-07	Log in with invalid information.	1. Open Login screen. 2. Enter email or password invalid. 3. Click Register.	System rejects login and shows invalid information message. Session is not created.	Customer is on Login screen.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Login Rejected
TC-ANH-AUTH-08	Log in with wrong password.	1. Open Login screen. 2. Enter valid email but wrong password. 3. Click Login.	System rejects login and shows an invalid credentials message. Session is not created.	A valid account exists.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Login Rejected
TC-ANH-AUTH-09	Log in with a locked account.	1. Open Login screen. 2. Enter credentials of a locked account. 3. Click Login.	System rejects login and shows account locked/status message.	The selected account is locked by Admin/Staff.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Login Rejected
TC-ANH-AUTH-10	Login with missing required fields.	1. Open Login screen. 2. Leave email or password empty. 3. Click Login.	System rejects login and shows required fields does not empty message. Session is not created.	Customer is on Login screen.	Failed	22/07/2020	Nguyễn Ngọc Anh	Passed	##	Nguyễn Ngọc Anh	Actual: Empty Inputs Allowed
TC-ANH-AUTH-11	Log out from the system.	1. Log in successfully. 2. Click Logout. 3. Try to access a protected URL again.	System destroys session and redirects to Login/Home page. Protected URL requires login again.	User is logged in.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Logout Destroyed
Scenario 3 - Forgot Password											
TC-ANH-AUTH-12	Password reset successful with valid email address	1. Open the Forgot Password screen. 2. Enter your registered email address. 3. Click Submit/Send.	The system verifies that the email exists, sends the email, and notifies of success.	The customer is not logged in; the email address already exists in the system.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			
TC-ANH-AUTH-13	Forgot password with non-existent email address	1. Open the Forgot Password screen. 2. Enter a non-registered email address. 3. Click Submit.	The system reports an error "Email does not exist in the system" and fails to send the email.	The customer is not logged in; the email address does not exist in the system.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			
Scenario 4 - Change Password											
TC-ANH-AUTH-14	Password change successful when the old password is entered correctly	1. Log in to the system. 2. Go to the Change Password screen. 3. Enter your current password, a valid new password, and confirm the new password. 4. Click Save/Update.	The system successfully updated the new password and displayed a message.	User is logged in.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			
TC-ANH-AUTH-15	Password change failed because the current password is incorrect	1. Go to the Change Password screen. 2. Enter the wrong current password. 3. Enter a valid new password and confirm the new password. 4. Click Save.	The system reported an error "Current password is incorrect" and did not change the password.	User is logged in.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			
TC-ANH-AUTH-16	Password change failed because the new password is not a duplicate	1. Go to the Change Password screen. 2. Enter the correct current password. 3. Enter a different new password and confirm the new password. 4. Click Save.	The system reported an error "Password confirmation does not match" and did not change the password.	User is logged in.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			
Scenario 5 - Customer Profile											
TC-ANH-PROF-01	Update customer profile with valid personal information.	1. Log in as Customer. 2. Open Profile screen. 3. Update full name, phone, address, ID number and driver license information. 4. Save.	System saves updated profile and displays the latest information correctly.	Customer is logged in.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Address updated successfully
TC-ANH-PROF-02	Update profile with invalid phone number format.	1. Log in as Customer. 2. Open Profile screen. 3. Enter invalid phone number. 4. Click Save.	System shows phone format validation message and does not save invalid data.	Customer is logged in.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Invalid phone format
TC-ANH-PROF-03	Verify customer profile completeness before booking.	1. Log in as Staff/Admin. 2. Open customer profile verification. 3. Review ID card and driver license fields. 4. Mark profile as verified.	System changes profile verification status and allows the customer to continue booking when all required documents are valid.	Customer profile contains required ID and driver license information.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Customer profile status
TC-ANH-PROF-04	Reject customer profile verification when required document information is missing.	1. Log in as Staff/Admin. 2. Open incomplete customer profile. 3. Try to verify the profile.	System prevents verification or records rejection reason. Customer status remains unverified.	Customer profile is missing ID or driver license information.	Passed	22/07/2020	Nguyễn Ngọc Anh	N/A			Actual: Verification Rejected

Final Status (after Round 1 & 2)

Workflow	Internal User Account Management Workflow			
Test requirement	Covers Admin/Staff user management functions: view user list, search/filter users, update account information and lock/unlock user account.			
Number of TCs	6			
Testing Round	Passed	Failed	Pending	N/A
Round 1	6	0	0	0
Round 2	0	0	0	6
Final	6	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note	Final Status (after Round 1 & 2)
Scenario 4 - View and Search Users												
TC-ANH-USER-01	Admin views the user list.	1. Log in as Admin. 2. Open User Management. 3. Observe the user list.	System displays user list with name, email/phone, role and account status.	Admin account exists and user data exists.	Passed	22/07/2026	Nguyễn Ngọc Anh	N/A			Actual: Loaded list of	Passed
TC-ANH-USER-02	Search user by keyword.	1. Open User Management. 2. Enter keyword by name/email/phone. 3. Click Search.	System displays users matching the keyword and shows no unrelated records.	Admin/Staff is logged in; matching data exists.	Passed	22/07/2026	Nguyễn Ngọc Anh	N/A			Actual: Found matching	Passed
TC-ANH-USER-03	Filter users by role or status.	1. Open User Management. 2. Select role/status filter. 3. Apply filter.	System displays users matching selected role/status only.	User data contains multiple roles/statuses.	Passed	22/07/2026	Nguyễn Ngọc Anh	N/A			Actual: Users filtered by	Passed
Scenario 5 - Update and Lock Users												
TC-ANH-USER-04	Update user account information successfully.	1. Log in as Admin. 2. Open a user detail. 3. Edit allowed fields such as name, role or status. 4. Save.	System saves the updated account and shows success message.	Admin is logged in; target account exists.	Passed	22/07/2026	Nguyễn Ngọc Anh	N/A			Actual: User account details	Passed
TC-ANH-USER-05	Lock a user account.	1. Log in as Admin. 2. Open User Management. 3. Select active user. 4. Click Lock and confirm.	System changes account status to Locked and prevents that user from logging in.	Target user is active.	Passed	22/07/2026	Nguyễn Ngọc Anh	N/A			Actual: User account locked	Passed
TC-ANH-USER-06	Unlock a user account.	1. Log in as Admin. 2. Open User Management. 3. Select locked user. 4. Click Unlock and confirm.	System changes account status to Active and allows login again.	Target user is locked.	Passed	22/07/2026	Nguyễn Ngọc Anh	N/A			Actual: User account unlocked	Passed

Workflow		Staff Booking Review, Calendar and Policy Workflow			
Test requirements					
Number of TCs		7			
Testing Round		Passed	Failed	Pending	N/A
Round 1		7	0	0	0
Round 2		0	0	0	0
Final		7	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note	Final Status (after Round 1 & 2)
Scenario 1 - Staff Reviews Calendar and Approves a Booking (Happy Path)												
TC-BAC-CAL-01	View booking calendar as Staff/Admin.	1. Log in as Staff/Admin. 2. Open Booking Calendar. 3. Select month/week view.	System displays bookings by vehicle and date so staff can avoid schedule conflicts.	Staff/Admin is logged in; booking data exists.	Passed	22/07/2024	Bui Xuan BAC	N/A			Actual: Booking calendar	Passed
TC-BAC-CAL-02	Filter booking calendar by vehicle/status/date range.	1. Open Booking Calendar. 2. Select vehicle/status/date filter. 3. Apply filter.	System displays matching bookings only and keeps calendar readable.	Booking data exists.	Passed	22/07/2024	Bui Xuan BAC	N/A			Actual: Filtered calendar	Passed
TC-BAC-POL-01	View booking policy before processing booking.	1. Log in as Staff/Admin. 2. Open Booking Policy/Booking Management. 3. Review deposit and cancellation rules.	System displays configured booking policy used when processing booking.	Booking policy data exists.	Passed	22/07/2024	Bui Xuan BAC	N/A			Actual: Booking policy	Passed
TC-BAC-PROC-01	Staff approves a valid pending booking.	1. Log in as Staff.	System changes booking status to Confirmed and prepares the next contract/deposit step.	Pending booking exists; no schedule	Passed	22/07/2024	Bui Xuan BAC	N/A			Actual:	Passed
Scenario 2 - Staff Rejects a Booking Request												
TC-BAC-PROC-02	Staff rejects a booking with a reason.	1. Open pending booking detail. 2. Click Reject. 3. Enter rejection reason. 4. Confirm.	System changes booking status to Rejected and stores rejection reason for customer notification.	Pending booking exists.	Passed	22/07/2024	Bui Xuan BAC	N/A			Actual: Rejected booking successful	Passed
Scenario 3 - Staff Cannot Approve Booking Due to Vehicle Conflict												
TC-BAC-PROC-03	Reject approving booking when vehicle is no longer available.	1. Open pending booking detail. 2. Simulate another approved overlapping booking.	System prevents approval and shows vehicle unavailable/conflict message.	Pending booking overlaps with an already approved booking.	Passed	22/07/2024	Bui Xuan BAC	N/A			Actual: Approved	Passed
Scenario 4 - Unauthorized User Attempts to Access Staff Booking Interface												
TC-BAC-PROC-04	Request unauthorized user from accessing bookings.	1. Log in as Customer.	System denies access and shows error/message denied or redirects to allowed area.	Customer account exists.	Passed	22/07/2024	Bui Xuan BAC	N/A			Actual:	Passed

Workflow	Customer Booking Creation and Management Workflow			
Test requirement	14			
Number of TCs	14			
Testing Round				
Round 1	Passed	Failed	Pending	N/A
Round 2	14	0	0	0
Round 3	0	0	0	14
Final	14	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note	Final Status (after Round 1 & 2)
Scenario 1 - Customer Books a Vehicle Successfully (Happy Path)												
TC-BAC-AVL-01	Check availability for a vehicle with valid rental dates.	1. Log in as Customer. 2. Open vehicle detail. 3. Select rental date and return date. 4. Click Check Availability.	System confirms the vehicle is available for the selected date range.	Vehicle is active and has no overlapping approved booking.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Car is available during booking	Passed
TC-BAC-BOOK-03	Create a booking request successfully.	1. Log in as Customer. 2. Select available vehicle and valid dates. 3. Fill pickup/return location and notes if required.	System creates a booking with Pending status and displays booking information to the customer.	Customer has completed required profile information; vehicle is available.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Booking created	Passed
TC-BAC-BOOK-04	Create booking and verify default status.	1. Create booking successfully. 2. Open My Bookings. 3. Check the newly created booking status.	New booking appears in My Bookings with Pending status.	Booking creation succeeds.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Initial status	Passed
TC-BAC-MYBK-01	View booking list as Customer.	1. Log in as Customer. 2. Open My Bookings.	System displays only bookings belonging to the logged-in customer.	Customer has at least one booking.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Retrieved #	Passed
TC-BAC-MYBK-02	View booking detail.	1. Open My Bookings. 2. Select a booking. 3. Open detail.	System displays vehicle, customer, date, price/deposit and booking status correctly.	Selected booking exists.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Viewing details	Passed
Scenario 2 - Customer Cannot Check Availability (Negative / Validation Paths)												
TC-BAC-AVL-02	Check availability when vehicle is already booked in the selected period.	1. Select a vehicle with an approved booking. 2. Enter overlapping rental and return dates. 3. Click Check Availability.	System shows unavailable message and prevents booking creation for the overlapping period.	Vehicle has an existing approved booking in the same period.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Car correct	Passed
TC-BAC-AVL-03	Check availability with return date before rental date.	1. Open check availability form. 2. Enter return date before rental date.	System shows invalid date validation and does not continue.	Customer is logged in.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Rejects	Passed
TC-BAC-AVL-04	Check availability without logging in.	1. Open vehicle detail as Guest. 2. Select rental dates. 3. Click booking/check availability action.	System allows viewing public information but requires login before creating booking.	Guest is not logged in.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Guest allowed	Passed
Scenario 3 - Customer Cannot Create Booking (Validation / Business Rule Paths)												
TC-BAC-BOOK-02	Reject booking when required booking information is missing.	1. Open Create Booking. 2. Leave required fields empty.	System highlights missing fields and booking is not created.	Customer is logged in.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Rejects	Passed
TC-BAC-BOOK-03	Reject booking when customer profile is not verified/complete.	1. Log in as Customer with incomplete profile. 2. Try to create booking.	System blocks booking or asks customer to complete/verify profile first.	Customer profile is incomplete or unverified.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Rejects #	Passed
Scenario 4 - Customer Updates a Pending Booking												
TC-BAC-UPD-01	Update a pending booking with valid new dates.	1. Open pending booking detail. 2. Click Update.	System updates booking information and keeps correct status/history according to business rule.	Booking status is Pending and vehicle is available for new dates.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Booking	Passed
TC-BAC-UPD-02	Open approved/handled-over booking detail.	1. Open approved/handled-over booking detail. 2. Try to update date or vehicle.	System prevents update and shows message that modification is not allowed in current status.	Booking status is Confirmed, Contracted or Handover/Returned.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Rejects #	Passed
Scenario 5 - Customer Cancels a Pending Booking												
TC-BAC-CAN-01	Cancel a pending booking.	1. Open pending booking detail.	System changes booking status to Cancelled and releases vehicle availability for selected period.	Booking status is Pending.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Booking	Passed
TC-BAC-CAN-02	Reject cancellation when booking is already handled.	1. Open handled-over booking detail.	System prevents cancellation and shows cancellation not allowed message.	Booking is in handover/rental process.	Passed	23/07/2026	Bui Xuân Bắc	N/A			Actual: Booking	Passed

Workflow	Vehicle Handover, Return Processing and Deposit Settlement Workflow			
Test requirement	Covers staff operational workflow after booking approval: process handover, process return, settle deposit and update operational status.			
Number of TCs	10			
Testing Round	Passed	Failed	Pending	N/A
Round 1	10	0	0	0
Round 2	0	0	0	10
Final	10	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note
Scenario 1 - Vehicle											
TC-TAM-HAND-01	Process vehicle handover successfully.	1. Log in as Staff. 2. Open Process Handover. 3. Select approved contract/booking. 4. Enter odometer, fuel level, vehicle condition and handover note. 5. Save.	System creates a vehicle handover record. Booking status is updated to In Progress. Vehicle status is updated to Rented.	Approved contract exists; deposit requirement is satisfied according to policy.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Handover process Passed
TC-TAM-HAND-02	Reject handover when deposit is not paid.	1. Open Process Handover for approved booking. 2. Confirm deposit has not been paid. 3. Try to complete handover.	System prevents handover and shows deposit/payment required message.	Approved booking exists but deposit is unpaid.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Handover Passed
TC-TAM-HAND-03	Reject handover with missing odometer/fuel/condition information.	1. Open Process Handover. 2. Leave odometer, fuel or condition empty. 3. Save.	System highlights required fields and handover is not completed.	Staff is logged in; approved contract exists.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Handover Passed
TC-TAM-HAND-04	Prevent duplicate handover for the same booking.	1. Complete handover for a booking. 2. Try to process handover again for the same booking.	System prevents duplicate handover and shows current status message.	Booking is already handed over.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Blocked Passed
Scenario 2 - Return											
TC-TAM-PRET-01	Process vehicle return with full information.	1. Log in as Staff. 2. Open Process Return. 3. Select active rental. 4. Enter return odometer, fuel level, return condition and notes. 5. Save.	System creates a vehicle return record. Booking status is updated according to payment status. Vehicle status is updated to Available or Maintenance.	Vehicle is currently handed over/active rental.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Return process Passed
TC-TAM-PRET-02	Record returned vehicle with damage note.	1. Open Process Return. 2. Select active rental. 3. Enter damage description and estimated fee. 4. Save.	System records damage information and creates/updates additional fee for settlement.	Vehicle is currently handed over/active rental.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Car return record Passed
TC-TAM-PRET-03	Reject return when return odometer is lower than handover odometer.	1. Open Process Return. 2. Enter return odometer lower than handover odometer. 3. Save.	System shows odometer validation and does not complete return.	Handover odometer exists.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Return entry Passed
Scenario 3 - Deposit											
TC-TAM-DEP-01	Settle deposit with no extra charge.	1. Open Settle Deposit after return.	System marks deposit as fully refundable/settled and closes settlement record.	Vehicle return completed; no extra fee exists.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Settled Passed
TC-TAM-DEP-02	Settle deposit with deducted additional fee.	1. Open Settle Deposit.	System deducts fee from deposit and displays remaining refundable/receivable amount correctly.	Return completed and additional fee exists.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Settled Passed
TC-TAM-DEP-03	Reject settlement before vehicle return is	1. Open Settle Deposit for active rental.	System prevents settlement and shows return completion required message.	Booking is handed over but not returned.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Blocked Passed

Final Status (after Round 1 & 2)

Workflow	Additional Fee and Reporting Workflow			
Test requirement	Covers additional fee recording and management reports such as revenue report and vehicle utilization report.			
Number of TCs	7			
Testing Round	Passed	Failed	Pending	N/A
Round 1	6	1	0	0
Round 2	1	0	0	0
Final	7	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note
Scenario 4 - Additional Fee											
TC-TAM-FEE-01	Record an additional fee after return.	1. Log in as Staff/Admin. 2. Open Record Additional Fee. 3. Select returned booking. 4. Enter fee type, amount and reason. 5. Save.	System records additional fee and links it to the booking/settlement.	Returned booking exists.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Late fee transaction recorded.
TC-TAM-FEE-02	Reject additional fee with invalid amount.	1. Open Record Additional Fee. 2. Enter zero/negative/non-numeric amount. 3. Save.	System shows amount validation and does not save invalid fee.	Staff/Admin is logged in.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Rejected recordin
TC-TAM-FEE-03	Update additional fee reason before settlement is completed.	1. Open existing additional fee. 2. Edit reason/description. 3. Save.	System saves updated fee reason while keeping audit information consistent.	Additional fee exists and settlement is not closed.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Addition al fee
Scenario 5 - Reports											
TC-TAM-REP-01	View revenue report by date range.	1. Log in as Admin/Staff. 2. Open Revenue Report. 3. Select valid date range. 4. Generate report.	System displays total revenue, deposits, rental payments and additional fees for selected date range.	Payment and booking data exists.	Failed	###	Trần Thị Minh Tâm	Passed		###	Trần Thị Minh Tâm Actual: Report loaded all
TC-TAM-REP-02	Export or view revenue report details.	1. Generate revenue report. 2. Open detail/export action if available.	System shows detailed rows or exports report data correctly.	Revenue report has data.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Detailed
TC-TAM-REP-03	View vehicle utilization report.	1. Log in as Admin/Staff. 2. Open Vehicle Utilization Report. 3. Select period. 4. Generate report.	System displays utilization rate/number of bookings per vehicle and supports management decisions.	Booking and vehicle data exists.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Vehicle utilization report
TC-TAM-REP-04	Handle report with no data.	1. Open report screen. 2. Select a date range with no transactions. 3. Generate report.	System shows zero values/no data message without error.	No payment/booking records exist for selected range.	Passed	###	Trần Thị Minh Tâm	N/A			Actual: Renders no data

Final Status (after Round 1 & 2)

Passed
Passed
Passed
Passed
Passed
Passed
Passed

Workflow	Contract Management and Payment Workflow			
Test requirement	Covers contract management, electronic signature, payment recording by Cash or Bank Transfer and export VAT invoice.			
Number of TCs	26			
Testing Round	Passed	Failed	Pending	N/A
Round 1	25	1	0	0
Round 2	1	0	0	25
Final	26	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note	Final Status (after Round 1 & 2)
Scenario 1 - Prepare Contract												
TC-TUNG-CON-01	Prepare contract successfully from confirmed booking.	1. Login as Staff/Admin. 2. Open a CONFIRMED booking. 3. Click Prepare Contract.	Contract is created in DRAFT status, booking/customer/vehicle information is copied and contract detail page is displayed.	Booking status = CONFIRMED, customer verified, vehicle exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-02	Reject contract creation when booking is not CONFIRMED.	1. Open a booking with status PENDING/CANCELLED. 2. Click Prepare Contract.	System displays "A contract can only be prepared for a Confirmed booking." No contract is created.	Booking status = CONFIRMED.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-03	Reject contract creation when customer profile is not verified.	2. Open confirmed booking. 2. Customer profile is UNVERIFIED. 3. Prepare contract.	System displays verification error and does not create contract.	Booking confirmed, customer not verified.	Failed	22/07/2026	Nguyễn Lâm Tung	Passed	2026-07-22 0:00:00	Nguyễn Lâm Tung	Actual	Passed
TC-TUNG-CON-04	Reject contract creation when vehicle does not exist.	2. Open confirmed booking linked to missing vehicle. 2. Prepare contract.	System displays "Vehicle not found." No contract is created.	Vehicle record missing.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
Scenario 2 - View Contract												
TC-TUNG-CON-05	View contract detail successfully.	1. Login. 2. Open Contract Management. 3. Select a contract.	Contract detail including customer, vehicle, rental fee and deposit is displayed.	Contract exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-06	Redirect unauthorized user to login page.	1. Access Contract Management without login.	System redirects to login page.	User not logged in.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-07	Reject viewing a non-existing contract.	1. Open contract detail using invalid contract ID.	System displays contract not found message.	Invalid contract ID.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-08	Deny viewing another customer's contract.	1. Login as Customer. 2. Access another customer's contract.	HTTP 403 Forbidden is returned.	Customer does not own contract.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
Scenario 3 - Update Contract												
TC-TUNG-CON-09	Update draft contract successfully.	1. Login as Staff/Admin. 2. Open DRAFT contract. 3. Edit fields. 4. Save.	Contract is updated and audit log is recorded.	Contract status = DRAFT.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-10	Reject updating non-DRAFT contract.	1. Open ACTIVE/COMPLETED contract. 2. Click Edit.	System displays edit restriction message.	Contract status = DRAFT.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-11	Prevent customer from updating contract.	1. Login as Customer. 2. Access Edit Contract.	HTTP 403 Forbidden.	Customer account.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-CON-12	Reject invalid contract information.	1. Edit contract. 2. Enter invalid dates or negative amount. 3. Save.	Validation errors displayed and contract is not updated.	Draft contract exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
Scenario 4 - Electronic Signature												
TC-TUNG-SIG-01	Complete electronic signature successfully.	1. Staff sends signature request. 2. Customer signs. 3. Staff signs.	Contract becomes ACTIVE and signed document is stored.	Draft contract exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-SIG-02	Reject signature when identity verification fails.	1. Customer enters invalid OTP.	Signature is rejected and retry is requested.	OTP verification enabled.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
Scenario 5 - Record Payment												
TC-TUNG-PAY-01	Record payment successfully.	1. Login as Staff/Admin. 2. Open Payment page. 3. Submit payment.	Payment status becomes COMPLETED and payment history is updated.	Booking exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-PAY-02	Reject payment with invalid amount.	1. Enter amount < 0. 2. Submit.	Validation error displayed.	Booking exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-PAY-03	Reject payment for non-existing booking.	3. Open payment page using invalid booking ID.	Booking not found message displayed.	Invalid booking ID.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-PAY-04	Approve pending cash payment.	1. Open Booking Detail. 2. Click Approve on pending CASH payment.	Payment status changes to COMPLETED and recorded by is updated.	Pending cash payment exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
Scenario 6 - Bank Transfer Payment												
TC-TUNG-BANK-01	Create bank transfer payment successfully.	1. Customer selects BANK TRANSFER. 2. Confirm payment.	Payment is created and QR code is displayed.	Bank transfer enabled.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-BANK-02	Complete payment through webbook.	1. Customer transfers money. 2. Webhook is received.	Payment becomes COMPLETED automatically.	Pending payment exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-BANK-03	Reject invalid webbook.	1. Send webbook with invalid signature.	HTTP 401/400 returned and payment status remains unchanged.	Pending payment exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
Scenario 7 - Record Refund												
TC-TUNG-REF-01	Record refund successfully.	1. Open Payment page. 2. Select REFUND. 3. Confirm.	Refund payment is recorded as COMPLETED using CASH.	Refund amount exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-REF-02	Reject invalid refund amount.	1. Enter refund amount < 0.	Validation error displayed.	Refund transaction.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
Scenario 8 - Export VAT Invoice												
TC-TUNG-VAT-01	Export VAT invoice successfully.	1. Open completed contract. 2. Click Export VAT Invoice.	VAT invoice is generated and linked to contract.	Contract COMPLETED and fully paid.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-VAT-02	Reject VAT export for unpaid contract.	1. Open unpaid contract. 2. Export VAT Invoice.	System displays export and displays error.	Contract not fully paid.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed
TC-TUNG-VAT-03	Reject duplicate VAT invoice.	1. Export VAT invoice for contract that already has active invoice.	System rejects request with duplicate invoice message.	Active VAT invoice exists.	Passed	22/07/2026	Nguyễn Lâm Tung	N/A			Actual	Passed

Workflow	Policy Configuration Workflow			
Test requirement	Covers rental policy and payment configuration.			
Number of YCs	4			
Testing Round	Passed	Failed	Pending	N/A
Round 1	3	1	0	0
Round 2	1	0	0	3
Final	4	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note
Scenario 9 - Configure Payment											
TC-TUNG-SET-01	Update payment configuration successfully.	1. Login as Admin. 2. Edit payment settings. 3. Save.	Configuration is saved successfully.	Admin account.	Failed	22/07/2026	Nguyễn Lâm Tùng	Passed	###	Nguyễn	Actual: Passed
TC-TUNG-SET-02	Reject invalid payment configuration.	1. Enter invalid deposit percentage or missing bank info. 2. Save.	Validation messages displayed and configuration is not saved.	Admin account.	Passed	22/07/2026	Nguyễn Lâm Tùng	N/A		Nguyễn	Actual: Passed
Scenario 10 - Configure Rental Policy											
TC-TUNG-POL-01	Create new rental policy successfully.	1. Login as Admin. 2. Create new policy. 3. Save.	Policy is stored successfully.	Admin account.	Passed	22/07/2026	Nguyễn Lâm Tùng	N/A		Nguyễn	Actual: Passed
TC-TUNG-POL-02	Reject duplicate policy key.	1. Create policy using existing key.	Validation error displayed and policy is not created.	Existing policy key.	Passed	22/07/2026	Nguyễn Lâm Tùng	N/A		Nguyễn	Actual: Passed

Final Status (after Round 1 & 2)

Workflow	Vehicle Catalog, Search and Detail Workflow			
Test requirement	Covers customer-facing vehicle browsing functions, view vehicle catalog, search/filter vehicles and view vehicle detail before booking.			
Number of TCs	9			
Testing Round	Passed	Failed	Pending	N/A
Round 1	8	1	0	0
Round 2	1	0	0	8
Final	9	0	0	0

Passed
Failed
Pending
N/A

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note
Scenario 1 - Vehicle Catalog											
TC-TINH-CAT-01	View vehicle catalog as Guest.	1. Open Home/Vehicle Catalog without login. 2. Observe listed vehicles.	System displays available public vehicle information such as image, name, price, seat number and status.	Vehicle test data exists.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Found vehicle
TC-TINH-CAT-02	View vehicle catalog as Customer.	1. Log in as Customer. 2. Open Vehicle Catalog.	System displays vehicle list and customer actions such as view detail/check availability.	Customer is logged in; vehicle data exists.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Found vehicle
TC-TINH-CAT-03	Handle empty vehicle catalog.	1. Use test database with no active vehicles or set filter to no results. 2. Open Vehicle Catalog.	System shows a friendly empty-state/no result message and does not crash.	No active vehicle matches the current query.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Found vehicle
Scenario 2 - Search and Filter											
TC-TINH-SEARCH-01	Search vehicle by keyword.	1. Open Vehicle Catalog. 2. Enter vehicle brand/model keyword. 3. Click Search.	System displays vehicles whose brand/model/name matches the keyword.	Matching vehicle exists.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Found matching vehicle
TC-TINH-SEARCH-02	Filter vehicles by price range.	1. Open Vehicle Catalog. 2. Choose minimum and maximum rental price. 3. Apply filter.	System displays only vehicles within selected price range.	Vehicle data contains different prices.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Found vehicles
TC-TINH-SEARCH-03	Filter vehicles by availability/status.	1. Open Vehicle Catalog. 2. Select Available status. 3. Apply filter.	System displays available vehicles only and excludes rented/maintenance vehicles.	Vehicle data contains multiple statuses.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Found vehicles
TC-TINH-SEARCH-04	Search with no matching result.	1. Open Vehicle Catalog. 2. Enter a keyword that does not exist. 3. Click Search.	System shows no search result message and keeps search/filter controls usable.	No vehicle matches the keyword.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Return 0 results
Scenario 3 - Vehicle Detail											
TC-TINH-DET-01	View vehicle detail from catalog.	1. Open Vehicle Catalog.	System opens detail page and displays full vehicle information, price, deposit, description and status.	Selected vehicle exists.	Passed	22/07/2026	Hoàng Ngọc Tĩnh	N/A			Actual: Found vehicle
TC-TINH-DETAL-02	Prevent booking action for unavailable vehicle.	1. Open an unavailable/rented vehicle detail.	System disables booking action or shows unavailable message.	Selected vehicle status is unavailable/rented/maintenance.	Failed	22/07/2026	Hoàng Ngọc Tĩnh	Passed	##	##	Hoàng Actual: Found vehicle

Final Status (after Round 1 & 2)

Workflow	Project Management, Status and Monitoring, workflow			
Task completion	Project timeline, critical path management, timeline, milestones, status, progress, status, status, availability and critical path management			
Number of Vis				
History				
	Failed	Failed	Failed	OK
Event 1	1	1	1	1
Event 2	1	1	1	1
Event	1	1	1	1

[illegible]

Workflow		Vehicle Management, Status and Maintenance Workflow			
Test requirement		Covers internal vehicle management functions, create/update vehicle, manage vehicle status, monitor availability and schedule maintenance.			
Number of TCs		8			
Testing Round		Passed	Failed	Pending	N/A
Round 1		7	0	0	0
Round 2		7	0	0	0
Final		7	0	0	0

Test Case ID	Test Case Description	Test Case Procedure	Expected Results	Pre-conditions	Round 1	Test date	Tester	Round 2	Test date	Tester	Note	Final Status (after Round 1 & 2)
Scenario 4 - Vehicle Management												
TC-TINH-AUDIT-01	Admin views audit log list	1. Log in as Admin 2. Navigate to Audit Log page 3. View activity list	System displays audit log list in Vietnamese (Tạo xe, Cập nhật xe, Xóa xe, Ấn xe). Timestamps show relative to Admin account exists; audit log entries exist	Passed	22/07/2026	Hoàng Ngọc Tinh	Passed					Passed
TC-TINH-AUDIT-02	Filter audit logs by action & date	1. Log in as Admin 2. Open Audit Log page 3. Apply filters: Action=CREATE, Entity=Vehicle, Date Range=Last 7 days 4. Click Filter	Audit log list filtered to show only: action=CREATE, entity_type=VEHICLE, createdAt >= 7 days ago. Count of Admin logged in; audit entries with different	Passed	22/07/2026	Hoàng Ngọc Tinh	Passed					Passed
TC-TINH-AUDIT-03	View detailed activity information	1. Log in as Admin 2. Open Audit Log page 3. Click on audit entry row 4. View detail popup with Actor, Action, Entity, ID, Timestamp	Detail view displays: Hoàng Ngọc Tinh (STAFF) tạo mới xe Toyota Camry - Biển số 29A-12345 (H45). Timestamp Admin logged in; audit entry with rich description	Passed	22/07/2026	Hoàng Ngọc Tinh	Passed					Passed
TC-TINH-AUDIT-04	Audit logs record vehicle hide/show	1. Log in as Staff 2. Open Vehicle Management 3. Click Hide on AVAILABLE vehicle 4. Log in as Admin 5. Open Audit Log 6. View audit log entry created with action=HIDE, entity_type=VEHICLE. Description includes vehicle name and user	Staff account exists; vehicle with AVAILABLE	Passed	22/07/2026	Hoàng Ngọc Tinh	Passed					Passed
TC-TINH-AUDIT-05	Audit logs record vehicle creation	1. Log in as Staff 2. Open Vehicle Management 3. Click Add Vehicle 4. Enter vehicle details 5. Save 6. Admin logs in and opens Audit log entry created with action=CREATE, entity_type=VEHICLE. Description: Tạo mới xe [brand] [model]	Staff account exists; Admin account exists	Passed	22/07/2026	Hoàng Ngọc Tinh	Passed					Passed
TC-TINH-AUDIT-06	Audit logs record vehicle update	1. Log in as Staff 2. Open Vehicle Management 3. Select vehicle -- Edit 4. Update price/description/status 5. Save 6. Admin opens Audit log entry created with action=UPDATE, entity_type=VEHICLE. Description: Cập nhật xe [brand] [model]	Staff account exists; Admin account exists	Passed	22/07/2026	Hoàng Ngọc Tinh	Passed					Passed
TC-TINH-AUDIT-07	Permission check - Customer denied access	1. Log in as Staff 2. Perform vehicle action (create/update) 3. Log in as Customer 4. Try to access /audit-logs 5. Verify access denied	System denies access to Audit Log page for non-Admin users. Redirects to access-denied page or home page	Customer account exists and is logged in	Passed	22/07/2026	Hoàng Ngọc Tinh	Passed				Passed

Passed
Failed
Pending
N/A

Passed
Failed
Pending
N/A

ound 1 & 2)

Title	Description	Issue ID	URL	State	Assignee	Created At	Due Date	Milestone	Labels	Functions/Screens
Prepare system test cases for Customer Booking	Write positive, negative, validation and permission test cases for Customer Booking Creation and Management Workflow.	1		Closed	Bùi Xuân Bắc	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Booking Customer
Execute test cases for Customer Booking	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	2		Open	Bùi Xuân Bắc	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Booking Customer
Prepare system test cases for Staff Booking	Write positive, negative, validation and permission test cases for Staff Booking Review, Calendar and Policy Workflow.	3		Closed	Bùi Xuân Bắc	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Booking Staff Workflow
Execute test cases for Staff Booking Review,	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	4		Open	Bùi Xuân Bắc	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Booking Staff Workflow
Collect system test evidence	Save screenshots, notes and actual results after executing individual system test cases.	5		Open	Bùi Xuân Bắc	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Test Evidence
Prepare system test cases for Customer	Write positive, negative, validation and permission test cases for Customer Account, Login and Profile Workflow.	6		Closed	Nguyễn Ngọc Ánh	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Auth and Profile Workflow
Execute test cases for Customer Account, Login	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	7		Open	Nguyễn Ngọc Ánh	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Auth and Profile Workflow
Prepare system test cases for Internal User	Write positive, negative, validation and permission test cases for Internal User Account Management Workflow.	8		Closed	Nguyễn Ngọc Ánh	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	User Management
Execute test cases for Internal User Account	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	9		Open	Nguyễn Ngọc Ánh	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	User Management
Collect system test evidence	Save screenshots, notes and actual results after executing individual system test cases.	10		Open	Nguyễn Ngọc Ánh	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Test Evidence
Prepare system test cases for Vehicle Handover,	Write positive, negative, validation and permission test cases for Vehicle Handover, Return Processing and Deposit Settlement	11		Closed	Trần Thị Minh Tâm	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Handover Return
Execute test cases for Vehicle Handover, Return	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	12		Open	Trần Thị Minh Tâm	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Handover Return
Prepare system test cases for Additional Fee	Write positive, negative, validation and permission test cases for Additional Fee and Reporting Workflow.	13		Closed	Trần Thị Minh Tâm	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Report Fee Workflow
Execute test cases for Additional Fee and	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	14		Open	Trần Thị Minh Tâm	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Report Fee Workflow
Collect system test evidence	Save screenshots, notes and actual results after executing individual system test cases.	15		Open	Trần Thị Minh Tâm	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Test Evidence
Prepare system test cases for Contract and	Write positive, negative, validation, permission and business rule test cases for Contract, Electronic Signature, Payment, Bank	16		Closed	Nguyễn Lâm Tùng	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Contract and Payment
Execute test cases for Contract and Payment	Run Round 1 system test cases on the demo build, record actual results and attach screenshot evidence.	17		Open	Nguyễn Lâm Tùng	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Contract and Payment
Prepare system test cases for Policy	Write positive, negative and validation test cases for Payment Method Configuration and Rental Policy Configuration.	18		Closed	Nguyễn Lâm Tùng	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Policy Configuration
Execute test cases for Policy Configuration	Run Round 1 system test cases on the demo build, record actual results and attach screenshot evidence.	19		Open	Nguyễn Lâm Tùng	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Policy Configuration
Collect system test evidence	Collect screenshots, actual results, issue logs and testing notes after executing all system test cases.	20		Open	Nguyễn Lâm Tùng	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	All Modules
Prepare system test cases for Vehicle Catalog,	Write positive, negative, validation and permission test cases for Vehicle Catalog, Search and Detail Workflow.	21		Closed	Hoàng Ngọc Tĩnh	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Vehicle Customer
Execute test cases for Vehicle Catalog, Search	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	22		Open	Hoàng Ngọc Tĩnh	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Vehicle Customer
Prepare system test cases for Vehicle	Write positive, negative, validation and permission test cases for Vehicle Management, Status and Maintenance Workflow.	23		Closed	Hoàng Ngọc Tĩnh	10/06/2026	10/06/2026	Iteration 1	Task_3_Done	Vehicle Admin Workflow
Execute test cases for Vehicle Management,	Run Round 1 test cases on demo build and update Passed/Failed/Pending status with screenshot evidence.	24		Open	Hoàng Ngọc Tĩnh	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Vehicle Admin Workflow
Collect system test evidence	Save screenshots, notes and actual results after executing individual system test cases.	25		Open	Hoàng Ngọc Tĩnh	10/06/2026	11/06/2026	Iteration 1	Task_3_Doing	Test Evidence